1. Online protocol and benchmark

1. Full-information round order

There are N experts and T rounds. On round t:

1. commit using only earlier information;
2. reveal the full loss vector $\ell_t = (\ell_t(1), \ldots, \ell_t(N))$;
3. incur the learner's scalar loss $L_t(\mathcal{A})$;
4. use every coordinate of ℓ_t in the next update.

Full information means all expert losses are observed *after* commitment, not before it.

2. Non-anticipating adversary

The current ℓ_t may depend on the full past and the learner's public rule. If the learner uses a private current sample, however, ℓ_t must be fixed without observing that sample.

Thus an adversary may adapt to history and public probabilities, but may not wait for the learner's private draw and then choose the current losses.

3. External and sublinear regret

$$R_T = \sum_{t=1}^{T} L_t(\mathcal{A}) - \min_{i\in\{1,\ldots,N\}} \sum_{t=1}^{T} \ell_t(i).$$

The comparator is **one fixed expert**, selected in hindsight and evaluated on the same realized sequence. It is not a round-by-round switching oracle and does not estimate population risk.

$R_T = o(T)$, equivalently $\frac{R_T}{T} \to 0$, is **sublinear regret**: only the average excess loss relative to that fixed comparator vanishes. Neither absolute loss nor future i.i.d. performance is controlled.

2. Realizable binary experts

1. Halving algorithm

Start with all N binary experts active. Predict by majority vote, using a fixed tie rule. After a learner mistake, retain exactly the active experts that predicted the revealed label; after a correct prediction, leave the active set unchanged.

A **perfect expert** has zero loss on every round of the complete sequence. This realizability assumption is stronger than the existence of a best expert in hindsight.

2. Halving mistake bound

If at least one expert is perfect and Halving makes m_T mistakes, then each mistake removes at least half of the active set while the perfect expert survives:

$$1 \le |S_{T+1}| \le \frac{N}{2^{m_T}}, \quad m_T \le \log_2 N.$$

In this zero-one realizable setting the perfect comparator has loss 0, so external regret equals m_T. If the best expert makes even one mistake, the surviving-expert lower bound disappears; do not "add one" to the theorem.

3. Deterministic soft penalties

1. Deterministic Weighted Majority

Fix $\beta \in (0,1)$ and initialize $w_1(i) = 1$. Predict by current weighted majority with a fixed tie rule. After revealing the label,

$$w_{t+1}(i) = \begin{cases} \beta w_t(i) & \text{if expert } i \text{ is mistaken} \\ w_t(i) & \text{if expert } i \text{ is correct} \end{cases}.$$

All finite-round weights remain positive, so the empty-set failure of Halving is removed. Positivity alone does not imply additive sublinear regret.

2. Exact deterministic WM bound

Let m_T be the learner's mistakes and m_T^* the mistakes of the best fixed expert. With $W_t = \sum_i w_t(i)$, a learner mistake gives

$$W_{t+1} \le \frac{1+\beta}{2} W_t, \quad \beta^{m_T^*} \le W_{T+1}.$$

Therefore, for binary predictions and zero-one loss,

$$m_T \le \frac{\ln N + m_T^* \ln\left(\frac{1}{\beta}\right)}{\ln\left(\frac{2}{1+\beta}\right)}.$$

The denominator is positive. Direct division by $\ln\left(\frac{1+\beta}{2}\right) < 0$ requires reversing the inequality.

3. Deterministic limitation

With two constant experts predicting 0 and 1, an adversary can choose the opposite label after seeing any deterministic current prediction. Then

$$m_T = T, \quad m_T^* \le \frac{T}{2}, \quad R_T \ge \frac{T}{2}.$$

This respects choose-before-reveal and proves linear regret for deterministic zero-one learners. It does not contradict the exact WM theorem, whose m_T^* coefficient retains a factor-two-style limitation.

4. Randomize the current expert

1. Distribution, hidden draw, and mixture loss

Normalize the positive weights before seeing the current loss:

$$p_t(i) = \frac{w_t(i)}{\sum_j w_t(j)}, \quad I_t \sim p_t, \quad L_t = \sum_i p_t(i)\ell_t(i).$$

The public distribution p_t may be known, but the sampled expert I_t is private until ℓ_t is fixed. The realized loss is one coordinate $\ell_t(I_t)$; it need not equal the weighted average L_t.

2. Where expectation enters

Under the non-anticipating private-draw condition,

$$E[\ell_t(I_t) \mid \mathcal{F}_{t-1}, p_t, \ell_t] = \sum_i p_t(i)\ell_t(i) = L_t.$$

A deterministic mixture-loss theorem therefore becomes an expected sampled-loss theorem after summing and taking expectation. It is not a pathwise guarantee for every sequence of random draws.

3. Randomized Weighted Majority bound

For binary losses $\ell_t(i) \in \{0,1\}$ and $\beta \in [\frac{1}{2}, 1)$, use the same mistaken-expert penalty as deterministic WM. Let

$$L_T = \sum_t \sum_i p_t(i)\ell_t(i), \quad L_T^* = \min_i \sum_t \ell_t(i).$$

Then

$$L_T \le \frac{\ln N}{1-\beta} + (2-\beta)L_T^*,$$

$$L_T - L_T^* \le \frac{\ln N}{1-\beta} + (1-\beta)L_T^*.$$

For $N \ge 2$ and known T, choose $\beta = \max\left\{\frac{1}{2}, 1 - \sqrt{\ln\frac{N}{T}}\right\}$. Together with the trivial cap-active bound,

$$L_T - L_T^* \le 2\sqrt{T \ln N}.$$

For private sampling, read these as expected sampled-loss statements.

5. Linear multiplicative weights

1. Update and legal domain

For $n \ge 1$, initialize $w_1(i) = 1$ and set

$$\Phi_t = \sum_i w_t(i), \quad p_t(i) = \frac{w_t(i)}{\Phi_t},$$

$$w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i)).$$

The signed theorem assumes $\ell_t(i) \in [-1,1]$ and $0 < \eta \le \frac{1}{2}$. Hence every factor lies in $[1-\eta, 1+\eta] \subset (0,\infty)$. A negative cost increases weight. The strictly positive learning rate matters because the regret theorem divides by η.

2. Potential proof: learner route

Write the mixture cost as $L_t = p_t \cdot \ell_t$. The total weight satisfies the exact one-step identity

$$\Phi_{t+1} = \Phi_t(1 - \eta p_t \cdot \ell_t).$$

Using $1 - x \le e^{-x}$ and accumulating over rounds gives

$$\Phi_{T+1} \le n \exp\left(-\eta \sum_{t=1}^{T} p_t \cdot \ell_t\right).$$

The equality belongs to the linear factor $1 - \eta\ell_t(i)$; it is not the exponential expert update.

3. Potential proof: comparator route

For $x \in [-1,1]$ and $0 < \eta \le \frac{1}{2}$,

$$\ln(1-\eta x) \ge -\eta x - \eta^2|x|.$$

Thus every fixed comparator i supplies

$$\ln(\Phi_{T+1}) \ge \ln(w_{T+1}(i)) \ge -\eta\sum_t \ell_t(i) - \eta^2 \sum_t |\ell_t(i)|.$$

Keep the comparator-dependent magnitude term until the signed-cost theorem is complete; replacing it immediately by T loses the sharper result.

4. Linear MWU comparator theorem

Combining the two potential routes gives, for every fixed expert i,

$$\sum_{t=1}^{T} p_t \cdot \ell_t \le \sum_{t=1}^{T} \ell_t(i) + \eta \sum_{t=1}^{T} |\ell_t(i)| + \frac{\ln n}{\eta}.$$

For losses in $[0,1]$, comparison with the best fixed expert yields the coarser external-regret bound

$$R_T \le \eta T + \frac{\ln n}{\eta}.$$

If an expert is privately sampled from p_t, the learner side is expected cumulative loss under the same non-anticipating condition.

5. Fixed-horizon rate and edge cases

For $n \ge 2$ and known horizon, use the single fixed rate

$$\eta = \min\left\{\frac{1}{2}, \sqrt{\frac{\ln(n)}{T}}\right\}.$$

If the cap is active, combine the theorem with the trivial $R_T \le T$; do not claim uncapped square-root algebra. If $n = 1$, the learner and only expert coincide, so $R_T = 0$; choose any legal positive η rather than substituting $\sqrt{\frac{\ln(1)}{T}} = 0$. For unknown T, this proof supplies neither a time-varying schedule nor a doubling argument.

6. Keep the updates distinct

1. Algorithm identity is more than a proof pattern

1. **Linear MWU** updates online expert weights by $w_{t+1}(i) = w_t(i)(1 - \eta\ell_t(i))$.
2. **AdaBoost** updates fixed training-example weights by $D_{t+1}(j) \propto D_t(j)e^{-\alpha_t y_j h_t(x_j)}$ using signed margins.
3. A **gradient template** moves a decision vector additively: $x_{t+1} = x_t - \eta g_t$, before any feasibility correction.

Sharing a potential proof does not identify state, feedback, update, or theorem.

7. Choose by assumptions

1. Quick selection map

Situation	Use	Licensed conclusion
binary; perfect expert	Halving	$m_T \le \log_2 N$
binary; no realizability; deterministic	deterministic WM	exact β-dependent mistake comparison; not additive regret
binary; hidden private draw	randomized WM	mixture or expected sampled regret; not pathwise
$[0,1]$ losses; full feedback	linear MWU	fixed-comparator or external-regret bound

Check realizability, loss range, permission for private randomization, and the adversary's current information together. A best hindsight expert need not be perfect; a public p_t is not the private draw; mixture loss is an expectation, not every sampled outcome; and same-sequence regret is not population risk.